

plt_sim version 3.1

JLT, SAA

2025-06-23

History

- plt_sim version 3.1 Uses spawners_m8_beta.stan
- From plt_sim_v3.qmd 2025-06-23. This and prev use spawners_m8.stan
- From plt_sim_v2.qmd 2025-06-14
- From plt_sim.R 2025-05-29

Simulation

Goal: Automate multiple simulations to demonstrate annual values of *run size* and spawning parameters (*arrival_timing*, *arrival_spread*, *exit_lag*, *exit_spread*) are recovered by fitting model spawner_m8.stan. All four spawning params as groups (multi-level regression) of annual values.

Simulation parameters

Assign distributions to parameters for runs and spawner behaviour. Set years and days for surveys. Params logged or not as per JLT code: spawners_m8.stan

```
log_run_mu          <- 10 # log value for lognormal PDD,
log_run_sigma       <- 1  # log value for lognormal PDD,
arrival_timing_mu    <- 35 # normal
arrival_timing_sigma <- 5
arrival_spread_mu    <- 5  # lognormal
arrival_spread_sigma <- 1.35 # log value was 0.3
exit_lag_mu         <- 10  # lognormal
exit_lag_sigma      <- 1.35 # log value was 0.3
exit_spread_mu      <- 5   # lognormal
```

```

exit_spread_sigma      <- 1.35 # log value was 0.3
rnbinom_size           <- 10  # sampling error
years_n                <- 12  # 4x3 plot
days_n                <- 100 # checked below
pct_surveyed           <- 0.9 # runif frac days sampled
# derived
days      <- 1:days_n
years      <- 1:years_n

# check range for days. pnorm(2.326348) = 99%
k99 = qnorm(0.99,0,1)
day_first=arrival_timing_mu -
  k99*(arrival_spread_mu+k99*arrival_spread_sigma)
day_last=arrival_timing_mu +
  k99*(arrival_spread_mu+k99*arrival_spread_sigma) +
  exit_lag_mu +
  k99*(exit_lag_sigma+exit_spread_mu+k99*exit_spread_sigma)
cat("Range of days with spawners conceivably present:", round(day_first),'to',round(day_last),

```

Range of days with spawners conceivably present: 16 to 86

Draws for annual values

Draw random samples for annual runs and spawning params. Spawners present each day is cumulative arrived minus cumulative exited based on those draws.

```

run 95%CL: 3103 156367
draws: 35761 18317 17799 9781 8513 15701 31242 36775 37994 56100 4682 16095

arrival_timing 95%CL: 25.2 44.8
draws: 35.28 38.87 36.61 40.64 46.29 30.55 41.61 37.1 34.73 32.26 37.56 38.16

arrival_spread 95%CL: 2.78 9
draws: 5.06 6.95 3.63 4.03 4.08 7.79 8.67 3.24 3.74 3.61 4.38 4.85

exit_lag 95%CL: 5.55 18.01
draws: 22.56 9.77 18.13 10.63 8.78 8.49 11.04 10.92 8.26 12.15 13.45 8.52

exit_spread 95%CL: 2.78 9
draws: 4.22 3.49 3.93 3.77 3.71 5.67 5.51 7.57 6.44 6.26 3.02 7.82

```

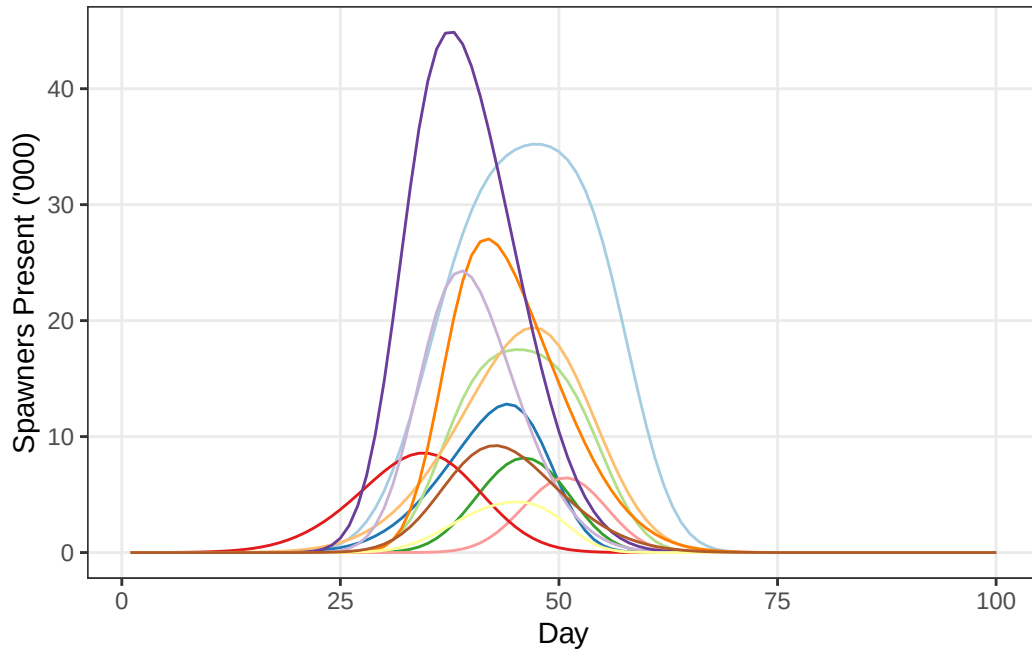


Figure 1: Time series of spawners present in each year.

Surveys

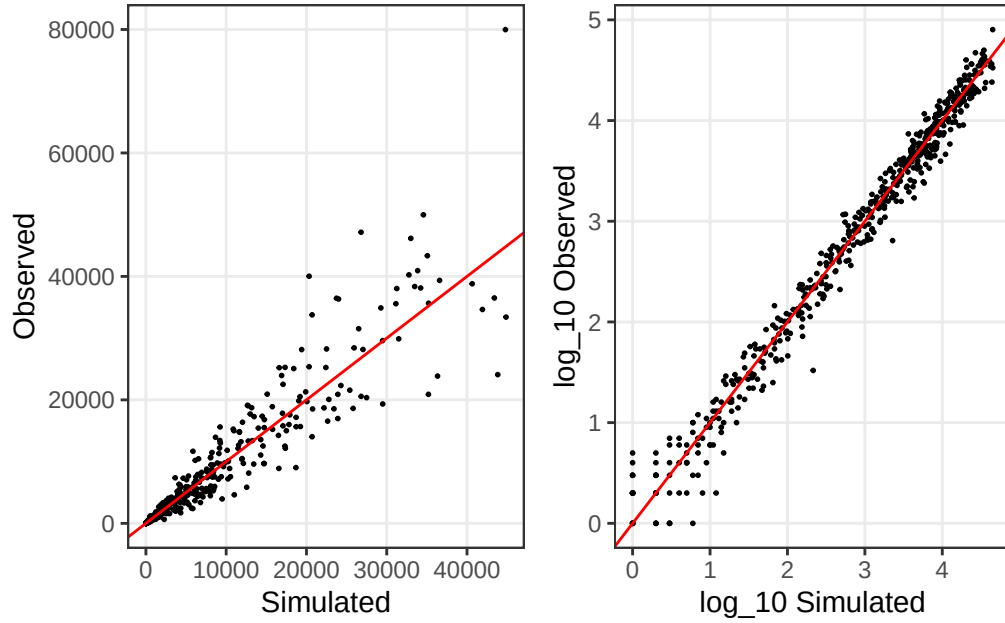
Define surveys as a random fraction of all days. Add negative binomial sampling error. Remove simulated values of zero and sampled observations of zero.

[1] max gap each year:

[1] 1 2 1 1 1 2 2 1 1 1 2 3

Observed time series

Plotted as observed to show scatter near peak (mode) day and as logs to better show negative skew when arrival_spread is greater than exit_spread (typical), and positive skew when smaller.



(a) Sampling error is negative binomial. Note $\log(0) = 1$.

Figure 2: Comparison of spawner observed to spawners present. Note $\log(0) = 1$.

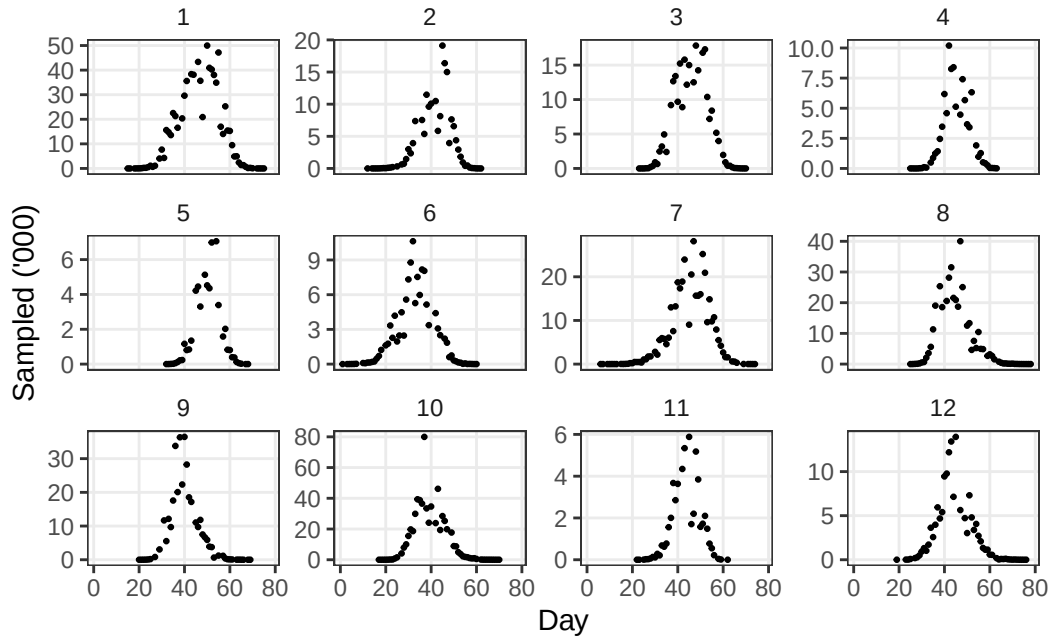
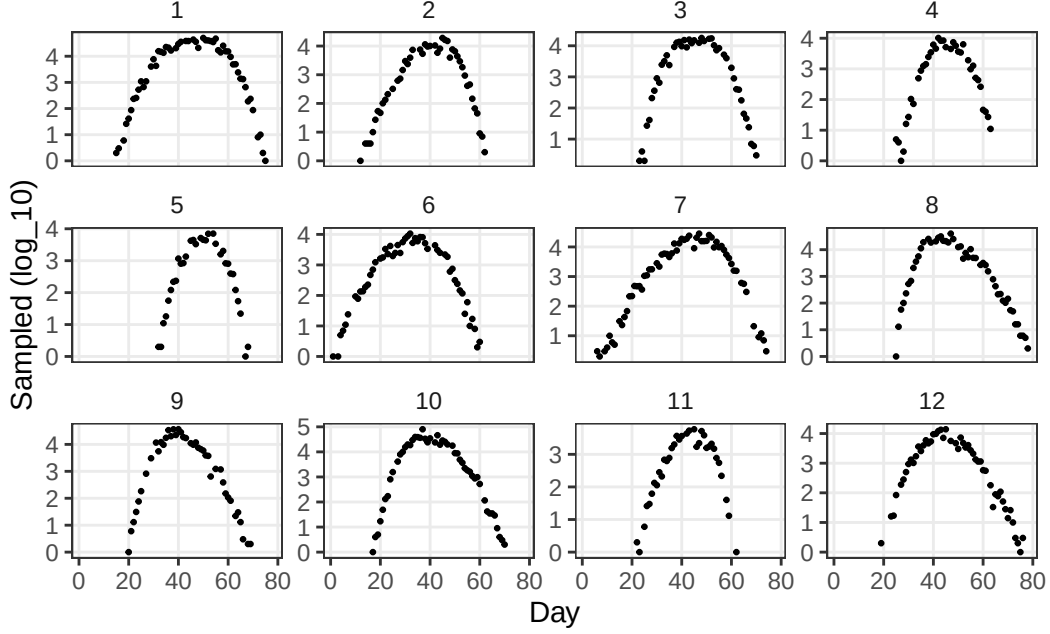


Figure 3: Annual time series of spawners observed.



(a) Reveals skewed distributions.

Figure 4: Annual time series, logged, of spawners observed.

Fit

Priors for Stan

Bayesian priors affect Markov Chain Monte Carlo (or related) sampling of the parameters hyperspace. The spawning parameters: *arrival_timing*, *arrival_spread*, *exit_lag*, *exit_spread* are treated as multilevel, groups of annual values. Priors for a group define an initial distribution for the annual values: $exit_lag = \Phi(\mu, \sigma)$ has group parameters *exit_lag_mu* and *exit_lag_sigma*. Annual values are sampled (fitted) from the group distribution, and group parameters are simultaneously fitted, thereby “sharing information” to improve the fit for each annual value. Note *log_runs* is not a group, the same prior applies to each year.

In what follows, the prior *exit_lag_sigma* uses the value for *spread_sigma*. The Stan code renames *count_dispersion* as $\log(live_phi)$ where *live_phi* is the size parameter for the negative binomial distribution.

Table 1: Priors for Stan sampling of parameters space.

prior	pdd	v1	v2
log_runs_mu	lnorm	10.000	1.0
arrival_mu	norm	40.000	5.0

Table 1: Priors for Stan sampling of parameters space.

prior	pdd	v1	v2
arrival_sigma	gamma	8.000	2.0
spread_mu	lnorm	1.946	0.3
spread_sigma	gamma	2.000	1.0
exit_lag_mu	lnorm	2.303	0.3
count_dispersion	lnorm	2.000	0.5

Data list for Stan

The priors are part of the Stan data list. Convergence of the HMC sampling chains is improved by providing initial values for *arrival_mu*. .

Stan sampling

Warning: There were 2 divergent transitions after warmup. See <https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup> to find out why this is a problem and how to eliminate them.

Warning: There were 6 transitions after warmup that exceeded the maximum treedepth. Increase <https://mc-stan.org/misc/warnings.html#maximum-treedepth-exceeded>

Warning: Examine the `pairs()` plot to diagnose sampling problems

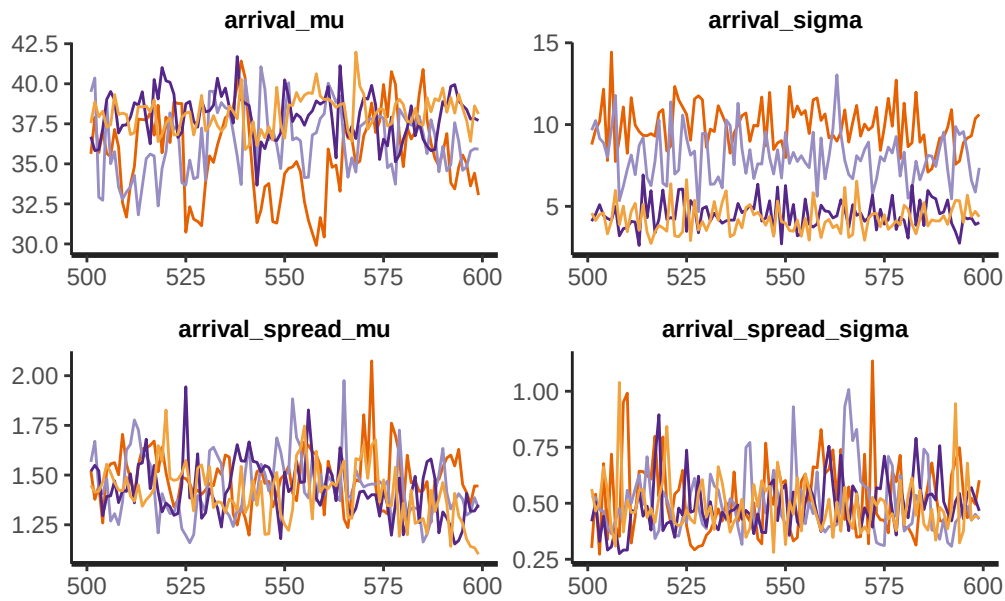
Warning: The largest R-hat is 2.43, indicating chains have not mixed. Running the chains for more iterations may help. See <https://mc-stan.org/misc/warnings.html#r-hat>

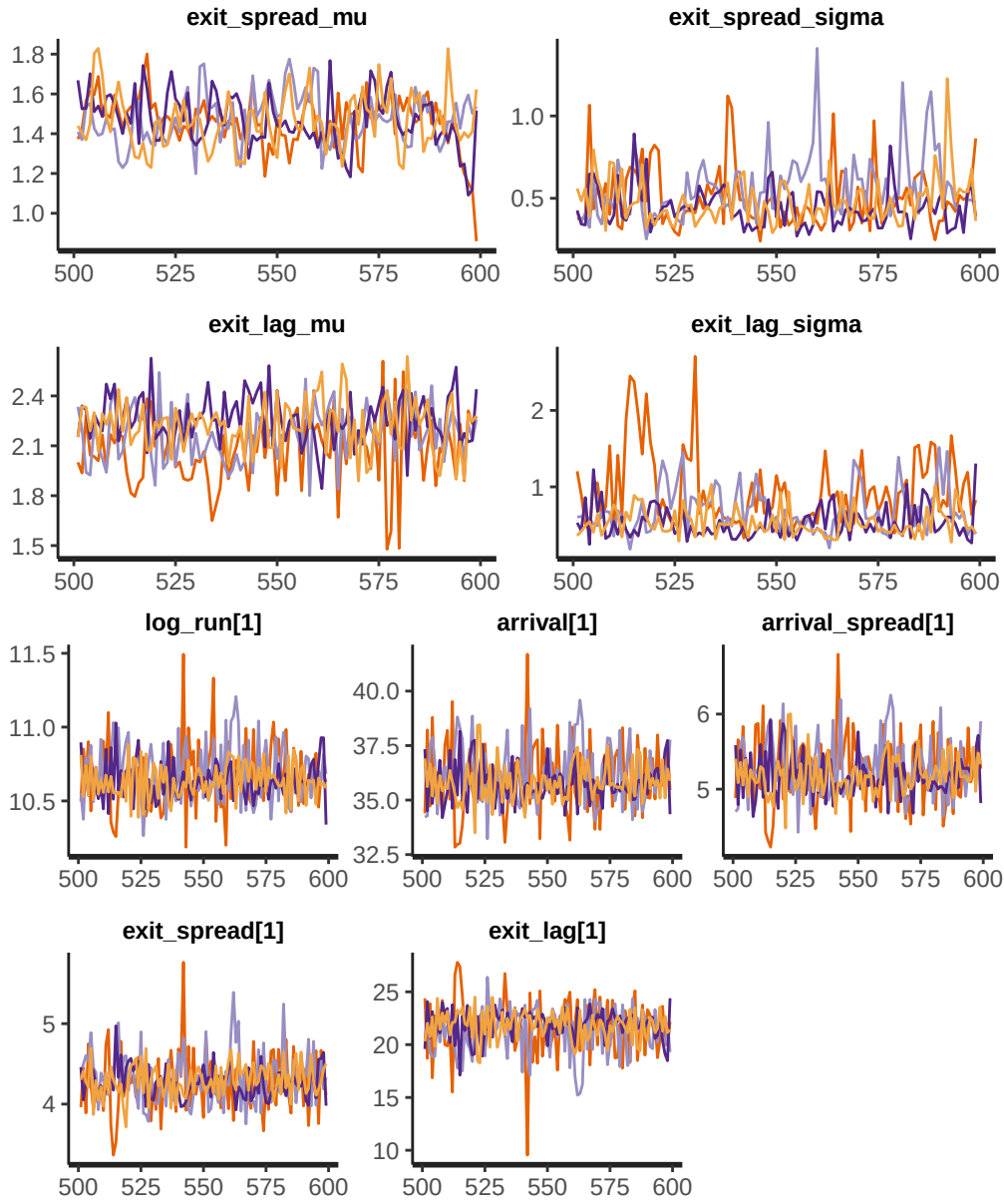
Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians. Running the chains for more iterations may help. See <https://mc-stan.org/misc/warnings.html#bulk-ess>

Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tails. Running the chains for more iterations may help. See <https://mc-stan.org/misc/warnings.html#tail-ess>

Table 2

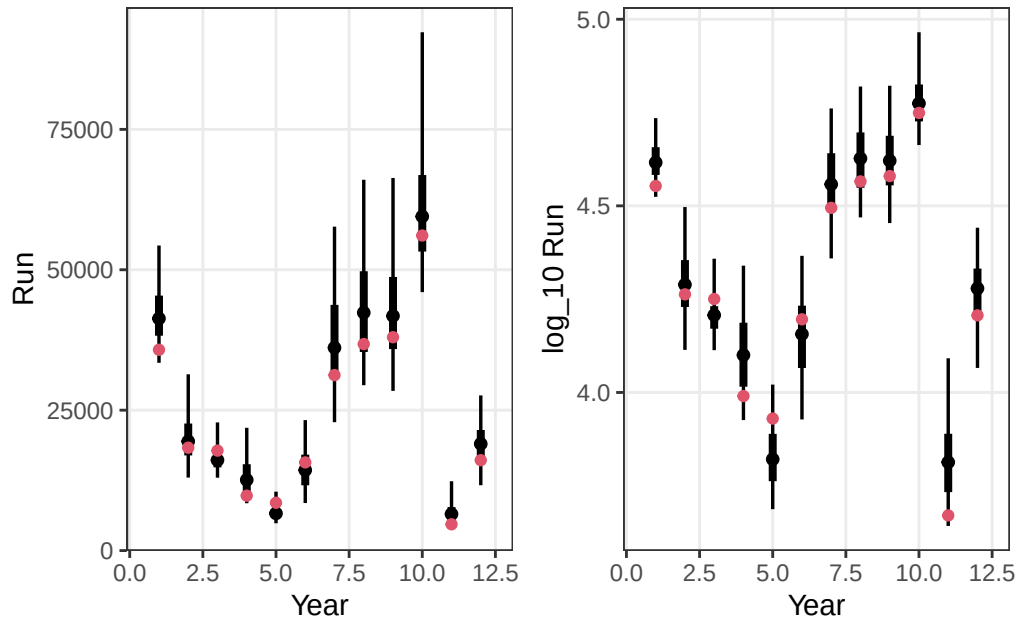
```
sp_dat = list(
  n_priors      = nrow(priors),
  priors        = data.matrix(priors[,3:4]),
  n_years       = years_n,
  n_obs         = nrow(sampled.df2),
  year          = sampled.df2$year,
  day           = sampled.df2$day,
  live_counts   = sampled.df2$sampled,
  live_phi      = rnbino_size,      # 10
  arrival_mu    = arrival_timing_mu ) # 35
```





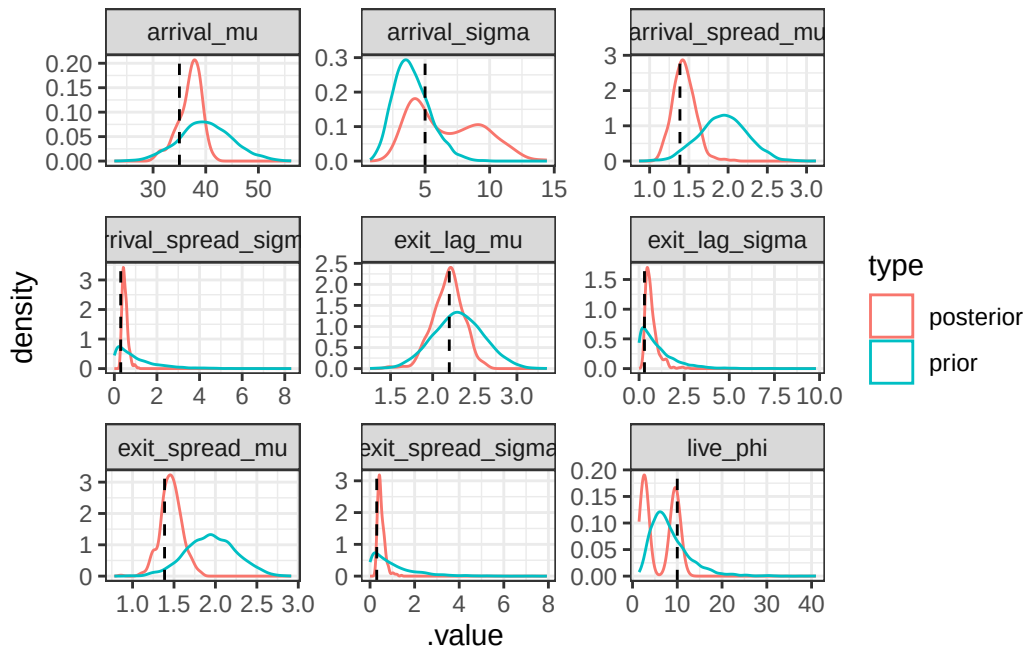
Fitted runs

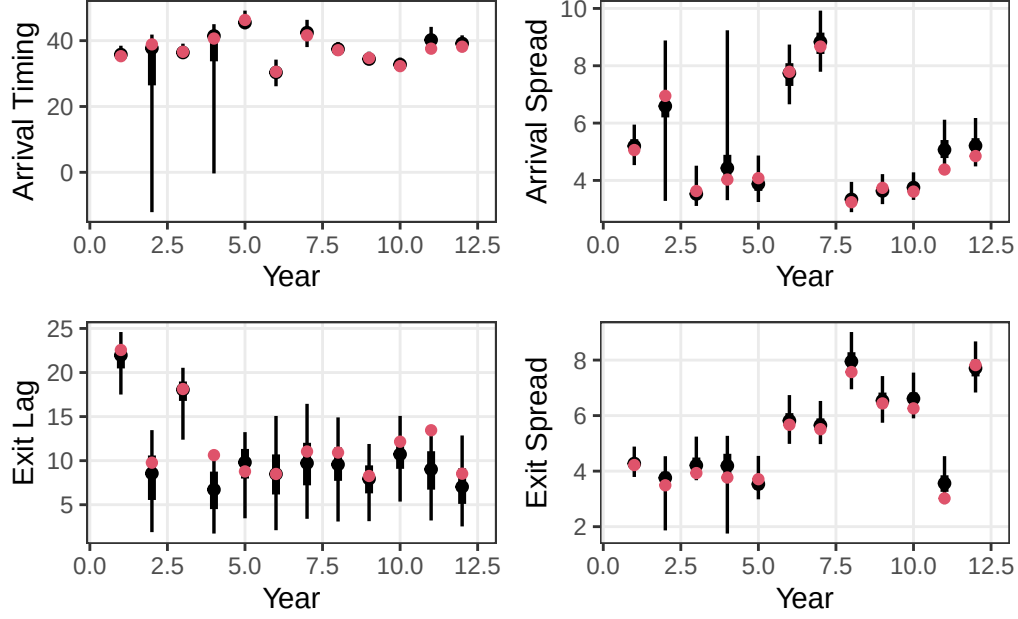
Spawner abundance estimates presented logged and arithmetic plots. $\log_{10} / \log_e = 0.4342945$



Fitted parameters

The annual values for spawner *arrival_timing*, *arrival_spread*, *exit_lag*, and *exit_spread* are presented as marginal posterior distributions (confidence limits) and compared to the simulated value. The goal is finding any consistent bias.





Spawner presence

The pattern of spawner distribution is recreated from the fitted parameters and presented with 95% confidence limits. This is compared to the observations with sampling error.

Stats

For comparison of scenarios of survey quality. Uses median absolute difference (MAD) from $\log(\text{observed}) - \log(\text{expected})$. It is faster to compute $\log(\text{observed}/\text{expected})$. The many samples representing the posterior PDD for observed are reduced to median. $\text{mad}(x)$ expands the value by 1.4826 to be the same as 1 stdev if x is a normal distribution. A mad of 0.16 is an accuracy of 16%, implying 95%CL is $\pm 2 \times MAD = 68\%$ to 132%.

Table 3: Statistics of annual parameters as ratios, each year's fitted median divided by the annual draw for that parameters before sampling error. Values for all years are summarized as median, to show bias, and mad (median absolute difference) to show precision.

Parameters	Median	MAD	CL95
Run	117%	7%	85% to 115%
Arrival Timing	101%	2%	95% to 105%
Arrival Spread	102%	8%	85% to 115%
Exit Lag	88%	13%	74% to 126%

Table 3: Statistics of annual parameters as ratios, each year’s fitted median divided by the annual draw for that parameters before sampling error. Values for all years are summarized as median, to show bias, and mad (median absolute difference) to show precision.

Parameters	Median	MAD	CL95
Exit Spread	104%	4%	91% to 109%

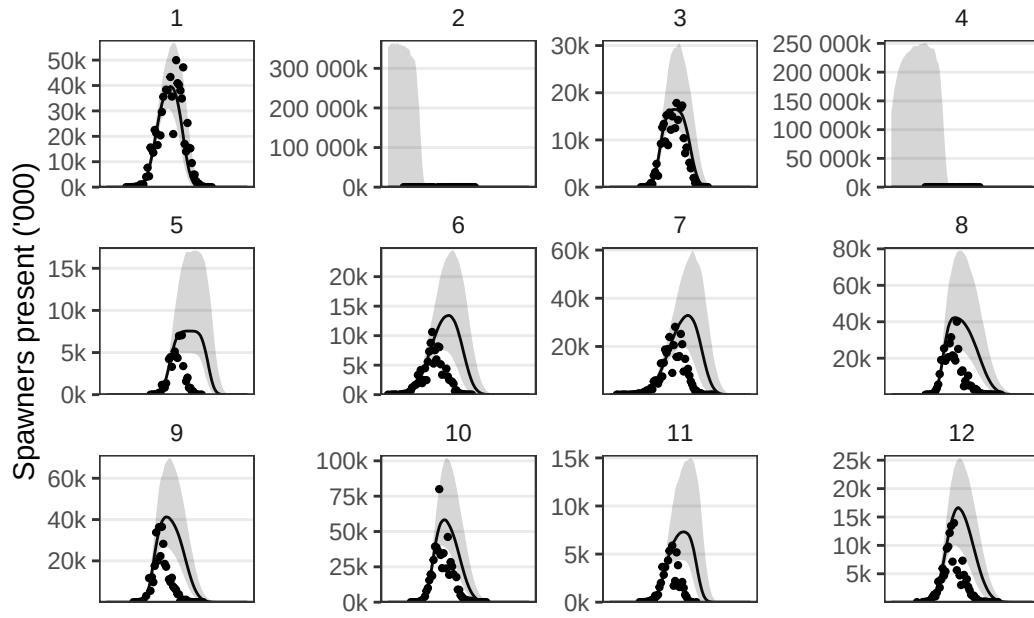


Figure 5: Estimated spawner presence, with 95% CL, from fitted parameters. Observed values are points.